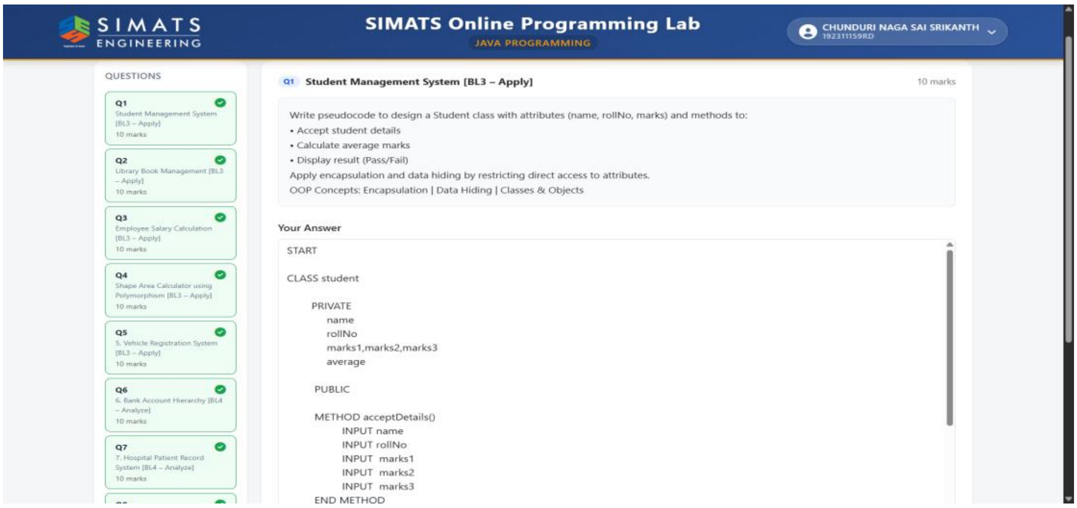
ASSESMENT TOOL 1 UNIT 1

1.)



2)

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Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

Your Answer

```
    METHOD issueBook()
        IF isAvailable =TRUE THEN
              isAvailable <-FALSE
              DISPLAY "Book Issued"
        ELSE
             DISPLAY "Book Not Available"
        END IF
    END METHOD

    METHOD returnBook()
         IF isAvailable =FALSE THEN
             isAvailable <-TRUE
             DISPLAY "Book Returned"
         ELSE
             DISPLAY "Book Already Available"
         END IF
    END METHOD

    METHOD displayBook()
           DISPLAY "Title : ",title
           DISPLAY "Author :",author
           DISPLAY "ISBN: ",ISBN
           DISPLAY "Available :",isAvailable
     END METHOD

END CLASS

CREATE  Book B1
CREATE  Book B2
CREATE  Book B3
```

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– Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
           DISPLAY "Author :",author
           DISPLAY "ISBN: ",ISBN
           DISPLAY "Available :",isAvailable
     END METHOD

END CLASS

CREATE  Book B1
CREATE  Book B2
CREATE  Book B3

CALL  B1.setBookDetails("java","james",101)
CALL  B2.setBookDetails("python","sri",201)
CALL  B3.setBookDetails("C++","baji",301)

CALL  B1.issueBook()
CALL  B2.issueBook()
CALL  B3.issueBook()

CALL  B1.displayBook()
CALL  B2.displayBook()
CALL  B3.displayBook()

STOP
```

Save Answer

3.

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q3 **Employee Salary Calculation [BL3 – Apply]** 10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to: •
Calculate gross salary (basic + HRA + DA)
• Calculate net salary after deductions (PF, tax)
• Display pay slip
Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
BEGIN

CLASS Employee
|
   PRIVATE
       empId
       name
       designation
       basicPay
       hra
       da
       pf
       tax
       grossSalary
       netSalary

   PUBLIC

   METHOD getEmployeeDetails()
```

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[BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

```
    METHOD getEmployeeDetails()

        INPUT empId
        INPUT name
        INPUT designation
        INPUT basicPay

    END METHOD


    METHOD calculateGrossSalary()

        hra ← basicPay × 20 / 100
        da ← basicPay × 10 / 100

        grossSalary ← basicPay + hra + da

    END METHOD


    METHOD calculateNetSalary()

        pf ← basicPay × 12 / 100

        tax ← grossSalary × 5 / 100

        netSalary ← grossSalary − (pf + tax)

    END METHOD


    METHOD displayPaySlip()
```

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```
    METHOD displayPaySlip()

        PRINT "---------- PAY SLIP ----------"

        PRINT "Employee ID      :", empId
        PRINT "Employee Name    :", name
        PRINT "Designation      :", designation

        PRINT "Basic Pay        :", basicPay
        PRINT "HRA              :", hra
        PRINT "DA               :", da

        PRINT "Gross Salary     :", grossSalary

        PRINT "PF Deduction     :", pf
        PRINT "Tax Deduction    :", tax

        PRINT "Net Salary       :", netSalary

        PRINT "------------------------------"

    END METHOD

END CLASS

DECLARE emp AS Employee

CALL emp.getEmployeeDetails()

CALL emp.calculateGrossSalary()

CALL emp.calculateNetSalary()

CALL emp.displayPaySlip()

END
```

10/10 answered

Save Answer

4)

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QUESTIONS

Q1 Student Management System [BL3 – Apply] 10 marks

Q2 Library Book Management [BL3 – Apply] 10 marks

Q3 Employee Salary Calculation [BL3 – Apply] 10 marks

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks

Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8

Q4 **Shape Area Calculator using Polymorphism [BL3 – Apply]** 10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.

OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
BEGIN

CLASS Shape
   METHOD calculateArea()
      DISPLAY "Area Calculation"
   END METHOD
END CLASS

CLASS Circle EXTENDS Shape
   DECLARE radius

   CONSTRUCTOR(radius)
      SET this.radius = radius
   END CONSTRUCTOR

   METHOD calculateArea()
      area = 3.14 × radius × radius
      DISPLAY "Area of Circle = ", area
   END METHOD
```


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Employee Salary Calculation [BL3 – Apply] 10 marks

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks

Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks

Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks

Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

Your Answer

```
END CLASS

CLASS Rectangle EXTENDS Shape
   DECLARE length, breadth

   CONSTRUCTOR(length, breadth)
      SET this.length = length
      SET this.breadth = breadth
   END CONSTRUCTOR

   METHOD calculateArea()
      area = length × breadth
      DISPLAY "Area of Rectangle = ", area
   END METHOD
END CLASS

CLASS Triangle EXTENDS Shape
   DECLARE base, height

   CONSTRUCTOR(base, height)
      SET this.base = base
      SET this.height = height
   END CONSTRUCTOR

   METHOD calculateArea()
      area = (base × height) / 2
      DISPLAY "Area of Triangle = ", area
   END METHOD
END CLASS

CREATE Shape reference s
```


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Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks

Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks

Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

10/10 answered

```
   CONSTRUCTOR(base, height)
      SET this.base = base
      SET this.height = height
   END CONSTRUCTOR

   METHOD calculateArea()
      area = (base × height) / 2
      DISPLAY "Area of Triangle = ", area
   END METHOD
END CLASS

CREATE Shape reference s

SET s = new Circle(7)
CALL s.calculateArea()

SET s = new Rectangle(10, 5)
CALL s.calculateArea()

SET s = new Triangle(8, 6)
CALL s.calculateArea()

END
```

Save Answer

5)

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QUESTIONS

Q1 Student Management System [BL3 – Apply] 10 marks

Q2 Library Book Management [BL3 – Apply] 10 marks

Q3 Employee Salary Calculation [BL3 – Apply] 10 marks

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks

Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8

Q5 **5. Vehicle Registration System [BL3 – Apply]** 10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
BEGIN

CLASS Vehicle

  DECLARE regNo
  DECLARE owner
  DECLARE fuelType

  METHOD displayDetails()
    DISPLAY "Registration Number :", regNo
    DISPLAY "Owner :", owner
    DISPLAY "Fuel Type :", fuelType
  END METHOD

END CLASS


CLASS TwoWheeler INHERITS Vehicle

  DECLARE engineCapacity
```


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Employee Salary Calculation [BL3 – Apply] 10 marks

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks

Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks

Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks

Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

```
  DECLARE engineCapacity

  METHOD displayDetails()

    DISPLAY "----- Two Wheeler Details -----"
    DISPLAY "Registration Number :", regNo
    DISPLAY "Owner :", owner
    DISPLAY "Fuel Type :", fuelType
    DISPLAY "Engine Capacity :", engineCapacity

  END METHOD

END CLASS


CLASS FourWheeler INHERITS Vehicle

  DECLARE seatingCapacity
  DECLARE acStatus

  METHOD displayDetails()

    DISPLAY "----- Four Wheeler Details -----"
    DISPLAY "Registration Number :", regNo
    DISPLAY "Owner :", owner
    DISPLAY "Fuel Type :", fuelType
    DISPLAY "Seating Capacity :", seatingCapacity
    DISPLAY "AC Status :", acStatus

  END METHOD

END CLASS
```

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Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    END METHOD

END CLASS


CREATE OBJECT bike OF TwoWheeler

SET bike.regNo ← "AP39AB1234"
SET bike.owner ← "Srikanth"
SET bike.fuelType ← "Petrol"
SET bike.engineCapacity ← 150

CALL bike.displayDetails()


CREATE OBJECT car OF FourWheeler

SET car.regNo ← "TS09CD5678"
SET car.owner ← "Ravi"
SET car.fuelType ← "Diesel"
SET car.seatingCapacity ← 5
SET car.acStatus ← "Available"

CALL car.displayDetails()

END
```

Save Answer

6)

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Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog

Q6 **6. Bank Account Hierarchy [BL4 – Analyze]** 10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.

OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
BEGIN

CLASS BankAccount

    VARIABLE accountNumber
    VARIABLE holderName
    VARIABLE balance

    METHOD withdraw(amount)

        DISPLAY "Withdrawal"

    END METHOD

END CLASS


CLASS SavingsAccount EXTENDS BankAccount
```

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10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    VARIABLE interestRate

    METHOD withdraw(amount)

      IF amount <= balance THEN

        balance ← balance – amount

        DISPLAY "Withdrawal Successful"

      ELSE

        DISPLAY "Insufficient Balance"

      ENDIF

    END METHOD

END CLASS


CLASS CurrentAccount EXTENDS BankAccount

    VARIABLE overdraftLimit

    METHOD withdraw(amount)

      IF amount <= balance + overdraftLimit THEN

        balance ← balance – amount
```

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7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
        balance ← balance – amount

        DISPLAY "Withdrawal Successful"

      ELSE

        DISPLAY "Overdraft Limit Exceeded"

      ENDIF

    END METHOD

END CLASS

CREATE SavingsAccount

CREATE CurrentAccount

CALL SavingsAccount.withdraw(500)

CALL CurrentAccount.withdraw(1500)

DISPLAY SavingsAccount.balance

DISPLAY CurrentAccount.balance

END
```

Save Answer

7)

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QUESTIONS

Q1
Student Management System [BL3 – Apply]
10 marks

Q2
Library Book Management [BL3 – Apply]
10 marks

Q3
Employee Salary Calculation [BL3 – Apply]
10 marks

Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks

Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8

Q7 **7. Hospital Patient Record System [BL4 – Analyze]** 10 marks

Design a Patient base class and analyze how it extends into:
- InPatient — with ward number, admission date, and daily charges
- OutPatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
BEGIN

CLASS Patient

    PRIVATE diagnosis
    PRIVATE prescription

    VARIABLE patientName

    METHOD generateBill()

      RETURN 0

    END METHOD

END CLASS


CLASS InPatient EXTENDS Patient

    VARIABLE dailyCharge
```

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5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks

Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks

Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

10/10 answered

```
    RETURN dailyCharge × numberOfDays
  END METHOD
END CLASS

CLASS OutPatient EXTENDS Patient
  VARIABLE consultationFee
  METHOD generateBill()
    RETURN consultationFee
  END METHOD
END CLASS

CREATE InPatient
CREATE OutPatient
DISPLAY InPatient.generateBill()
DISPLAY OutPatient.generateBill()
END
```

Save Answer

8)

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QUESTIONS

Q1 Student Management System [BL3 – Apply] 10 marks

Q2 Library Book Management [BL3 – Apply] 10 marks

Q3 Employee Salary Calculation [BL3 – Apply] 10 marks

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks

Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8

Q8 **8. E-Commerce Product Catalog [BL4 – Analyze]** 10 marks

Analyze and design a Product class hierarchy for an e-commerce system:
- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
BEGIN
CLASS Product
  VARIABLE productName
  VARIABLE price
  METHOD applyDiscount()
    RETURN price
  END METHOD
END CLASS

CLASS Electronics EXTENDS Product
  METHOD applyDiscount()
    RETURN price – (price × 0.10)
```

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[BL3 – Apply] 10 marks

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks

Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks

Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks

Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks

Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks

Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks

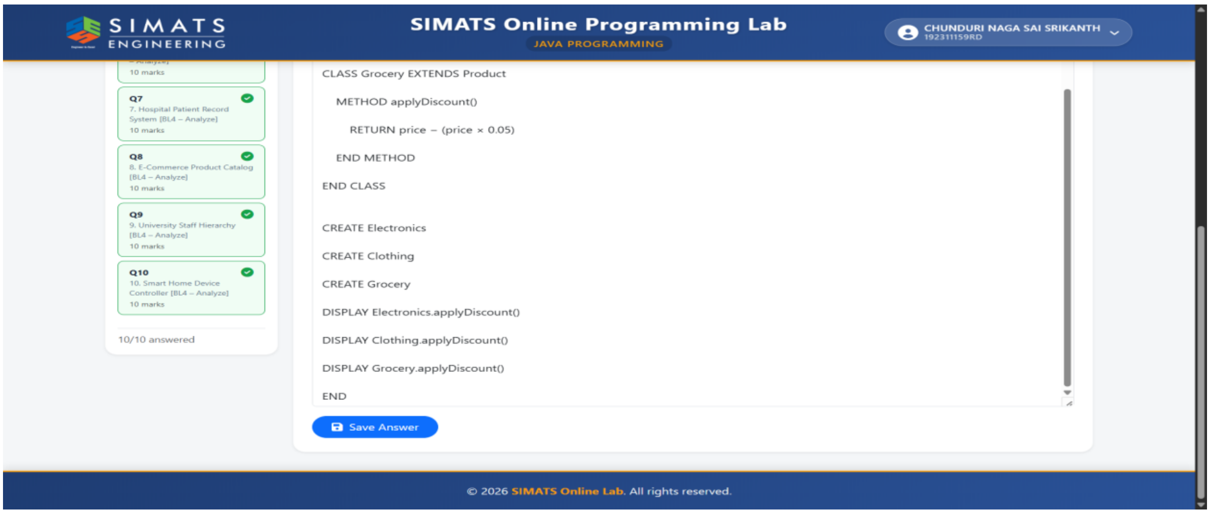
Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks

```
  END METHOD
END CLASS

CLASS Clothing EXTENDS Product
  METHOD applyDiscount()
    RETURN price – (price × 0.20)
  END METHOD
END CLASS

CLASS Grocery EXTENDS Product
  METHOD applyDiscount()
    RETURN price – (price × 0.05)
  END METHOD
END CLASS

CREATE Electronics
CREATE Clothing
```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10 marks
Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks
Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks
Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks
Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks
10/10 answered
CLASS Grocery EXTENDS Product
METHOD applyDiscount()
RETURN price − (price × 0.05)
END METHOD
END CLASS
CREATE Electronics
CREATE Clothing
CREATE Grocery
DISPLAY Electronics.applyDiscount()
DISPLAY Clothing.applyDiscount()
DISPLAY Grocery.applyDiscount()
END
Save Answer
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9)


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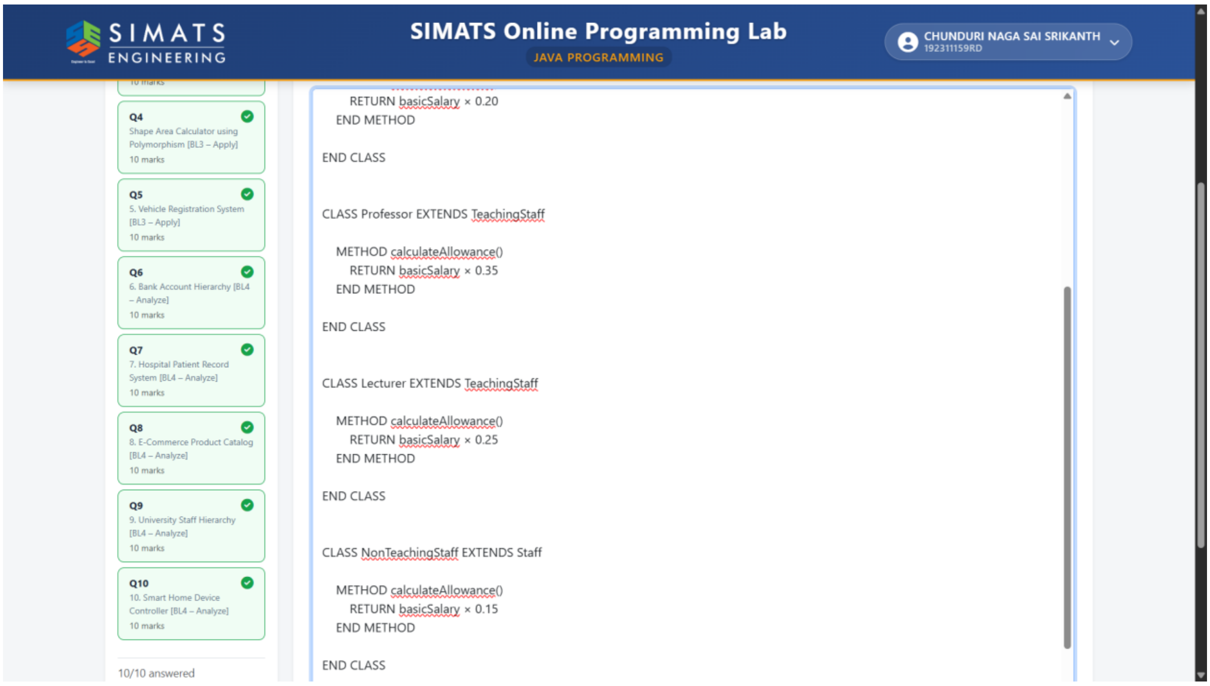
QUESTIONS
Q1
Student Management System [BL3 – Apply]
10 marks
Q2
Library Book Management [BL3 – Apply]
10 marks
Q3
Employee Salary Calculation [BL3 – Apply]
10 marks
Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks
Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks
Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks
Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks
Q8
Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks
Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:
• Identify which attributes should be private vs protected
• Override calculateAllowance() at each level
• Analyze how protected access enables inheritance without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.
OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding
Your Answer
BEGIN
CLASS Staff
PRIVATE staffID
PRIVATE name
PROTECTED basicSalary
METHOD calculateAllowance()
RETURN 0
END METHOD
END CLASS
CLASS TeachingStaff EXTENDS Staff
METHOD calculateAllowance()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10 marks
Q4
Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks
Q5
5. Vehicle Registration System [BL3 – Apply]
10 marks
Q6
6. Bank Account Hierarchy [BL4 – Analyze]
10 marks
Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks
Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks
Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks
Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks
10/10 answered
RETURN basicSalary × 0.20
END METHOD
END CLASS
CLASS Professor EXTENDS TeachingStaff
METHOD calculateAllowance()
RETURN basicSalary × 0.35
END METHOD
END CLASS
CLASS Lecturer EXTENDS TeachingStaff
METHOD calculateAllowance()
RETURN basicSalary × 0.25
END METHOD
END CLASS
CLASS NonTeachingStaff EXTENDS Staff
METHOD calculateAllowance()
RETURN basicSalary × 0.15
END METHOD
END CLASS

10 marks
Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks
Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks
Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks
Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks
10/10 answered

```
END CLASS

CLASS NonTeachingStaff EXTENDS Staff

   METHOD calculateAllowance()
      RETURN basicSalary × 0.15
   END METHOD

END CLASS


CREATE Professor
CREATE Lecturer
CREATE NonTeachingStaff

DISPLAY Professor.calculateAllowance()

DISPLAY Lecturer.calculateAllowance()

DISPLAY NonTeachingStaff.calculateAllowance()

END
```

Save Answer

10)

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QUESTIONS
Q1 Student Management System [BL3 – Apply] 10 marks
Q2 Library Book Management [BL3 – Apply] 10 marks
Q3 Employee Salary Calculation [BL3 – Apply] 10 marks
Q4 Shape Area Calculator using Polymorphism [BL3 – Apply] 10 marks
Q5 5. Vehicle Registration System [BL3 – Apply] 10 marks
Q6 6. Bank Account Hierarchy [BL4 – Analyze] 10 marks
Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks
Q8

Q10 **10. Smart Home Device Controller [BL4 – Analyze]** 10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:
- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
BEGIN

CLASS SmartDevice
   VARIABLE deviceName

   CONSTRUCTOR(name)
      deviceName ← name
   END CONSTRUCTOR

   METHOD executeCommand(command)
      DISPLAY "Generic Device"
   END METHOD
END CLASS


CLASS SmartLight EXTENDS SmartDevice

   METHOD executeCommand(command)
```

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6. Bank Account Hierarchy [BL4 – Analyze] 10 marks
Q7 7. Hospital Patient Record System [BL4 – Analyze] 10 marks
Q8 8. E-Commerce Product Catalog [BL4 – Analyze] 10 marks
Q9 9. University Staff Hierarchy [BL4 – Analyze] 10 marks
Q10 10. Smart Home Device Controller [BL4 – Analyze] 10 marks
10/10 answered

```
      IF command = "on" THEN
         DISPLAY "Light Switched ON"

      ELSE IF command = "off" THEN
         DISPLAY "Light Switched OFF"

      ELSE IF command = "status" THEN
         DISPLAY "Light is Working"

      ELSE
         DISPLAY "Invalid Command"

      ENDIF

   END METHOD

END CLASS


CLASS SmartThermostat EXTENDS SmartDevice

   METHOD executeCommand(command)

      IF command = "on" THEN
         DISPLAY "Thermostat Activated"

      ELSE IF command = "off" THEN
         DISPLAY "Thermostat Deactivated"

      ELSE IF command = "status" THEN
         DISPLAY "Temperature = 25°C"
```

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6. Bank Account Hierarchy [BL4 – Analyze]
10 marks

Q7
7. Hospital Patient Record System [BL4 – Analyze]
10 marks

Q8
8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    ELSE
        DISPLAY "Invalid Command"

    ENDIF

  END METHOD

END CLASS


CLASS SmartLock EXTENDS SmartDevice

  PRIVATE PIN

  METHOD executeCommand(command)

    IF command = "on" THEN
        DISPLAY "Door Locked"

    ELSE IF command = "off" THEN
        DISPLAY "Door Unlocked"

    ELSE IF command = "status" THEN
        DISPLAY "Lock Status Displayed"

    ELSE
        DISPLAY "Invalid Command"

    ENDIF

  END METHOD
```

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8. E-Commerce Product Catalog [BL4 – Analyze]
10 marks

Q9
9. University Staff Hierarchy [BL4 – Analyze]
10 marks

Q10
10. Smart Home Device Controller [BL4 – Analyze]
10 marks

10/10 answered

```
    ELSE IF command = "status" THEN
        DISPLAY "Lock Status Displayed"

    ELSE
        DISPLAY "Invalid Command"

    ENDIF

  END METHOD

END CLASS


CREATE Light
CREATE Thermostat
CREATE Lock

STORE all devices in DeviceList

FOR EACH device IN DeviceList
     CALL device.executeCommand("status")
END FOR

END
```

Save Answer